

Volume 09

Endpoint Security Engine (ESE)

Detection Layers

Layer 1

Configuration Assessment

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Layer 2

Software Inventory

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Layer 3

Process Inspection

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Layer 4

Persistence Analysis

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Layer 5

Network Analysis

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Layer 6

Behavior Analysis

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Layer 7

Evidence Correlation

↓

Layer 8

Threat Assessment

↓

Layer 9

Guided Response

New Core Engine

Endpoint Intelligence Engine (EIE)

This engine builds a security profile of every endpoint.

Supported platforms:

- Windows

- Ubuntu/Linux
- macOS

Future support:

- Android
 - iOS (within platform limitations)
-

Endpoint Digital Twin

Every monitored computer has a Digital Twin.

Example:

Windows Laptop

The Digital Twin lets us detect meaningful changes over time instead of looking only at the current state.

Compromise Assessment Engine (CAE)

One new idea.

Instead of asking:

"Is the PC hacked?"

We calculate:

Device Trust

Device Health

Exposure

Suspicious Activity

Persistence Indicators

Behavioral Indicators

Configuration Risk

Evidence Quality

The result is a transparent assessment rather than a binary yes/no answer.

Guided Investigation

If suspicious behavior is found:

Mission

Investigate My Computer

Project Sentinel automatically:

1. Collects evidence.
2. Reviews recent changes.
3. Checks startup mechanisms.
4. Reviews scheduled tasks.
5. Reviews services.
6. Reviews network activity.
7. Reviews local security settings.
8. Correlates findings.
9. Generates a report.

The user receives both an executive summary and detailed technical evidence.

Guided Remediation

If the platform determines that an item is suspicious but not confirmed malicious, it might recommend actions such as:

- Disable automatic startup.
- Remove an unnecessary scheduled task.
- Reset a browser extension.
- Revoke an unused credential.
- Isolate the device from the network (with user approval).
- Export evidence for further investigation.

Potentially destructive actions—such as deleting files or removing software—should only occur after user approval and, ideally, after backups or recovery points are available.

PS-ADR-0006

Endpoint Trust Assessment

Status: Accepted

Decision

Project Sentinel shall assess endpoint security using an evidence-based trust model. The platform shall not claim compromise without sufficient supporting evidence.

Rationale

Evidence-based assessments improve user trust, reduce false positives, and provide a defensible basis for remediation recommendations.

A strategic idea that could make Project Sentinel unique

I would like to add a long-term capability called the **Security Baseline Engine**.

Instead of relying only on signatures or one-time scans, Project Sentinel learns the **authorized and expected state** of each device:

- Expected operating system version.
- Approved software inventory.
- Expected services.
- Normal startup entries.
- Typical network behavior.
- Approved administrator accounts.
- Expected security settings.

Future assessments compare the current state against this baseline. That allows the platform to detect unexpected changes—such as a new startup item, an unknown service, or a configuration drift—even if no known malware signature exists. This is technically achievable, complements traditional detection methods, and fits perfectly with our object-based, evidence-first architecture.

=====[USER]=====

Yes and important that we give the user a uniq system that is easy to use and understand with output data. Good and continue

=====[ASSISTANT]=====

I completely agree, and I think this should become one of the defining principles of Project Sentinel.

Many cybersecurity products are powerful, but they overwhelm users with dashboards, CVEs, CVSS scores, logs, and technical jargon. Users are often left wondering:

"What should I actually do?"

Project Sentinel should answer that question immediately.

PROJECT SENTINEL BLUEPRINT

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